1. Introduction

1.1 Purpose

The purpose of this document is to present the Software Requirements Specification (SRS) for AquaSense: IoT-Based Smart Water Monitoring & Management System. This document provides a detailed description of the system's functional and non-functional requirements, system features, and overall design objectives. It serves as a reference for developers, project guides, and evaluators to understand the system's behavior, expected performance, and constraints. The SRS ensures clarity in system development and helps in maintaining consistency throughout the project lifecycle.

1.2 Scope

AquaSense is an IoT-based system designed to monitor and manage water usage efficiently in environments such as homes, hostels, apartments, and institutions. The system focuses on multi-area water monitoring, where water consumption is tracked independently across different sections such as kitchens, bathrooms, drinking water points, and plantation areas.

The system collects real-time data using flow sensors and tank level sensors, processes it using a microcontroller (ESP32/ESP8266), and displays the information on a mobile/web dashboard. It also provides intelligent suggestions for optimizing water usage and includes a servo-controlled valve mechanism for both manual and automatic flow regulation.

The scope of AquaSense includes real-time monitoring, data visualization, smart control, and efficient water management, aiming to reduce wastage and promote sustainable usage practices.

2. Overall Description

2.1 Product Perspective

AquaSense is an IoT-based smart water monitoring and management system designed to function as an integrated solution for efficient water usage. The system combines hardware components such as water flow sensors, tank level sensors, a microcontroller (ESP32/ESP8266), and a servo-controlled valve with a software layer consisting of a mobile/web dashboard.

It operates as a standalone system but relies on network connectivity for real-time monitoring and user interaction. AquaSense fits into existing water supply systems without requiring major infrastructure changes.

2.2 Product Functions

- The major functions of AquaSense include:
- Monitoring real-time water flow at multiple usage points
- Tracking area-wise consumption (kitchen, bathroom, etc.)
- Measuring tank water level and supply availability
- Processing sensor data using a microcontroller
- Displaying system data on a dashboard interface
- Providing smart suggestions to optimize water usage
- Allowing manual control of water flow
- Supporting Auto Mode for automatic regulation using a servo valve

2.3 User Classes and Characteristics

1. User (Primary Actor)

- Basic technical knowledge
- Uses mobile/web dashboard
- Monitors water usage and applies suggestions
- Can manually control or enable Auto Mode

2. Sensor / Hardware (System Actor)

- Includes flow sensors, level sensors, microcontroller, and valve
- Automatically collects and transmits data
- Requires no human intervention during operation
- Works continuously in real-time

2.4 Operating Environment

- Hardware Platform: ESP32 / ESP8266 microcontroller
- Sensors: Water flow sensors (YF-S201), ultrasonic level sensor
- Network: Wi-Fi connectivity
- Software Platform: Blynk / Web dashboard
- Devices Supported: Smartphones, tablets, laptops
- Operating Systems: Android, iOS, Windows (via browser)

2.5 Design and Implementation Constraints

- Limited processing power of microcontroller
- Dependence on stable Wi-Fi connection
- Accuracy limitations of low-cost sensors
- Power supply constraints for continuous operation
- Budget limitations (low-cost implementation requirement)
- Physical installation constraints in existing plumbing systems

2.6 Assumptions and Dependencies

- Continuous availability of Wi-Fi network
- Proper installation of sensors at designated locations
- Stable power supply for the system
- Users have access to a smartphone or web interface
- Sensors provide reasonably accurate readings
- Water supply system is compatible with sensor integration

3. System Features

3.1 Real-Time Water Flow Monitoring

The system continuously monitors the water flow at different points using flow sensors. It measures both input and output flow rates to provide accurate real-time data.

3.2 Multi-Area Water Consumption Tracking

AquaSense tracks water usage across multiple areas such as:

- Kitchen
- Bathrooms
- Drinking water stations
- Plantation/utility lines

This enables area-wise analysis and identification of high-consumption zones.

3.3 Tank Water Level Monitoring

The system measures the water level in the storage tank using a level sensor. It helps in preventing:

- Tank overflow
- Sudden water shortages

3.4 Supply Availability Detection

The system detects whether water supply is available or not by analyzing input flow data and updates the status on the dashboard.

3.5 Data Processing and Analysis

The microcontroller (ESP8266/ESP32) processes real-time sensor data to:

- Calculate flow rates
- Analyze consumption patterns
- Generate meaningful insights

3.6 Smart Suggestion System

Based on the collected data, the system provides suggestions such as:

- Reducing flow in high-usage areas
- Optimizing water distribution
- Preventing wastage

3.7 Servo-Based Flow Control

A servo motor is used to control the valve opening:

- Adjusts water flow rate
- Enables efficient regulation
- Supports both manual and automatic control

3.8 Manual Control via Dashboard

Users can manually control the system through a mobile/web dashboard:

- Adjust valve position
- Monitor real-time data
- Apply suggested actions

3.9 Auto Mode Operation

In Auto Mode, the system automatically:

- Adjusts water flow
- Balances supply and demand
- Ensures efficient distribution without user intervention

3.10 Dashboard Visualization

The system provides a user-friendly dashboard displaying:

- Input flow rate
- Output flow (area-wise)
- Tank level
- Supply status
- Suggestions and system actions

3.11 Wireless Communication

The system uses Wi-Fi connectivity to transmit data from the microcontroller to the dashboard in real time.

3.12 Alert and Notification System

The system alerts users in critical conditions such as:

- Low water level
- Excessive usage
- No supply availability

4. External Interface Requirements

4.1 User Interface

The AquaSense system provides a user-friendly mobile/web dashboard interface for monitoring and controlling water usage.

- The interface displays real-time data such as:
  - Input water flow rate
  - Area-wise water consumption (kitchen, bathroom, etc.)
  - Tank water level
  - Water supply status

- The dashboard includes:
  - Graphs and visual indicators for easy understanding
  - Buttons for manual control of the valve
  - Toggle option for enabling/disabling Auto Mode

- The UI is designed to be:
  - Simple and intuitive
  - Accessible on smartphones and web browsers
  - Responsive with minimal latency

4.2 Hardware Interface

The system interacts with various hardware components:

- Microcontroller (ESP8266/ESP32) acts as the central processing unit
- Water Flow Sensors measure input and output water flow
- Ultrasonic Sensor measures tank water level
- Servo Motor controls the water valve for flow regulation

Connections:

- Sensors are connected via digital I/O pins
- Servo motor is controlled via PWM signal
- Power supply is provided through regulated DC source

4.3 Software Interface

The AquaSense system interfaces with:

- Blynk IoT Platform / Web Dashboard
  - For real-time data visualization
  - For remote control of system components

- Embedded Firmware (Arduino IDE)
  - Used to program the ESP8266/ESP32
  - Handles sensor data processing and logic implementation

- The system uses libraries for:
  - Wi-Fi connectivity
  - Sensor data acquisition
  - Servo motor control

4.4 Communication Interface

The system uses wireless communication for data transfer:

- Wi-Fi Communication
  - Connects ESP8266/ESP32 to the internet
  - Transfers sensor data to the dashboard

- IoT Protocols
  - Uses HTTP/MQTT (depending on platform like Blynk)

- Data Transmission
  - Real-time updates sent at regular intervals
  - Bidirectional communication for:

I. Sending sensor data
II. Receiving user commands

- The system includes:
  - Auto-reconnect feature for network failure
  - Status indication for connectivity

5. Non-Functional Requirements

5.1 Performance Requirements

The system shall provide near real-time updates of water flow, tank level, and consumption data on the dashboard with minimal delay.

5.2 Reliability Requirements

The system shall operate continuously with stable performance and provide consistent and accurate readings from sensors.

5.3 Scalability Requirements

The system shall support the addition of multiple sensors and monitoring points (e.g., kitchen, bathroom, plantation) without significant redesign.

5.4 Usability Requirements

The dashboard interface shall be user-friendly, intuitive, and easy to understand for non-technical users.

5.5 Efficiency Requirements

The system shall consume minimal power and efficiently utilize hardware resources for long-term operation.

5.6 Security Requirements

The system shall restrict unauthorized access and ensure that only authorized users can view data and control the system.

5.7 Maintainability Requirements

The system shall be designed for easy maintenance, including sensor replacement and software updates.

5.8 Compatibility Requirements

The system shall be compatible with mobile devices and standard web browsers for dashboard access.

5.9 Accuracy Requirements

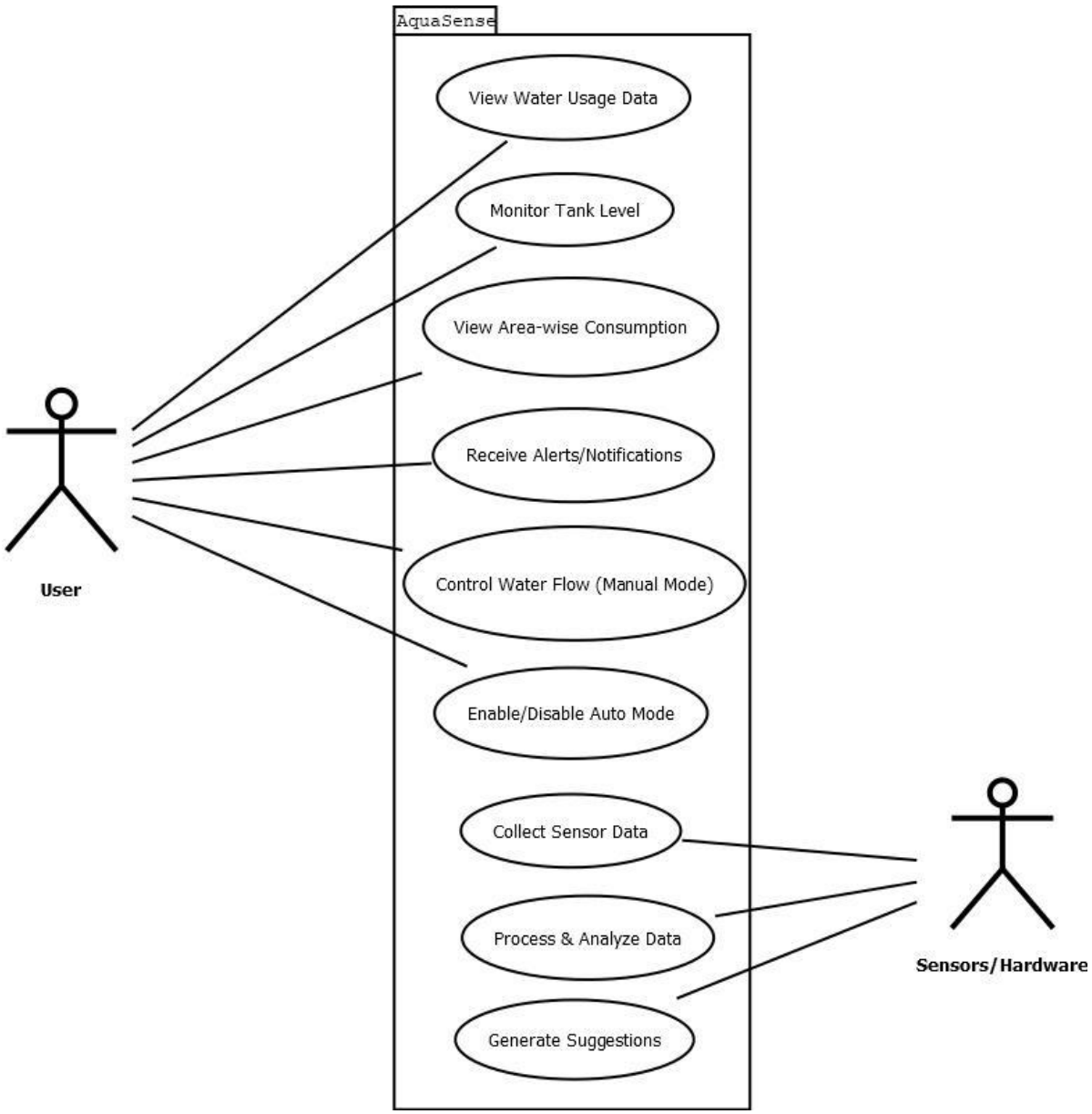
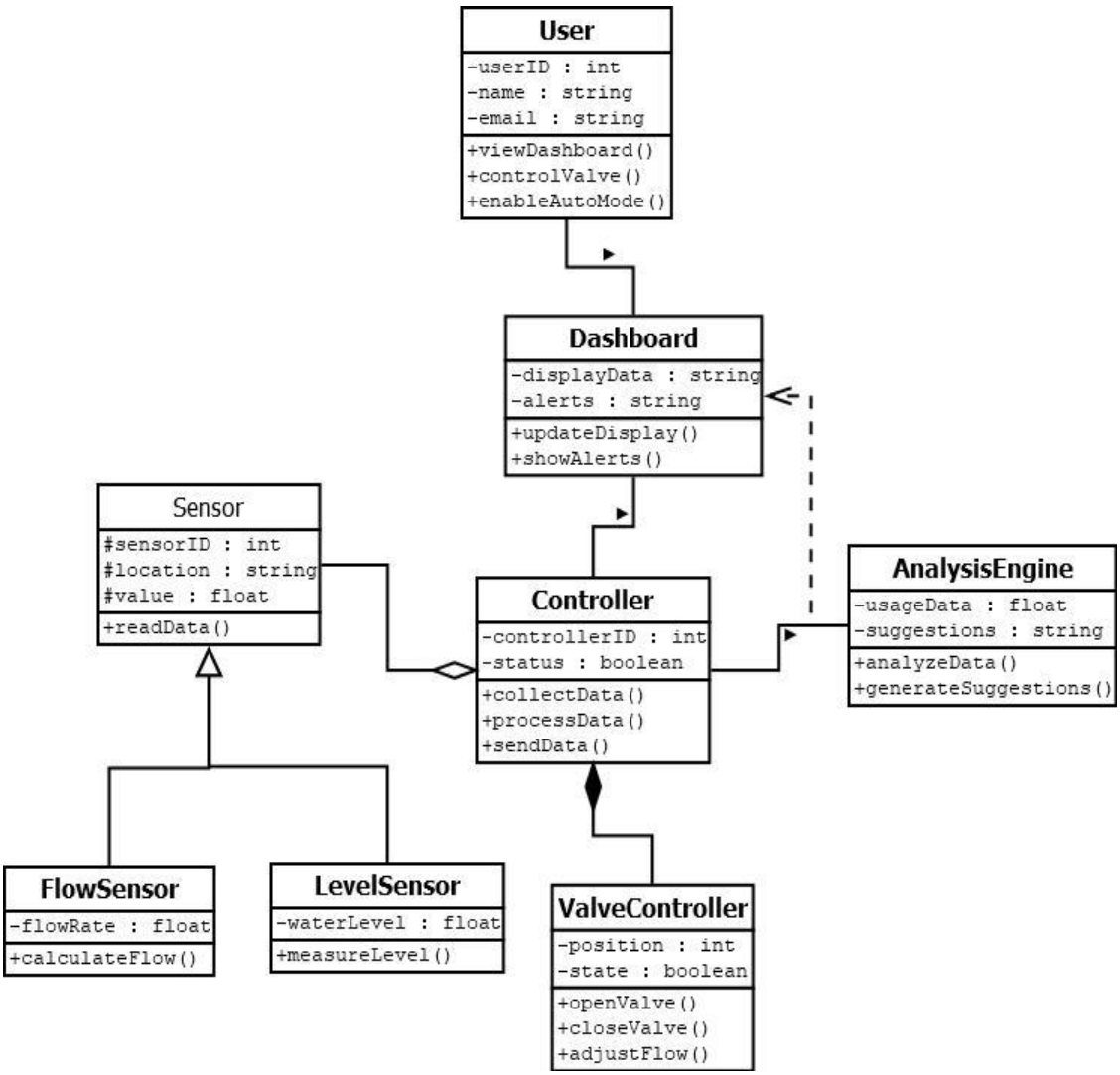
The system shall provide accurate measurements of water flow and tank levels with minimal error.

5.10 Availability Requirements

The system shall remain operational and accessible with minimal downtime, including stable Wi-Fi connectivity.

6. System Models

6.1 Use Case Diagram



6.2 Class Diagram